

2 (a) The gravitational potential at a point is defined as the work done per unit mass in bringing a small test mass from infinity to that point. B1

(b) (i) Increase in potential energy = final potential energy – initial potential energy

$$= -\frac{GMm}{r_2} - \left( -\frac{GMm}{r_1} \right)$$

B1

$$= GMm \left( \frac{1}{r_1} - \frac{1}{r_2} \right)$$

1 mark for correct potential energy formula (**WITH negative sign**) and substituted correctly.

(ii) **Work is done by thrusters**

M1

Hence **total** energy **increases / not constant**

M1

Increase in potential energy not equal to decrease in kinetic energy

A0

(c) Decrease in potential energy = increase of KE

$$0 - \left( -\frac{GM(m_r)}{r_2} \right) = \frac{1}{2}(m_r)v^2 - 0 \quad \text{or equate total energy}$$

M1 for correct decrease in potential energy, M1 for correct increase in KE

$$v = \sqrt{\frac{2GM}{r_2}}$$

A1 for the correct final expression for  $v$ .

